

Unifying Theory of Dynamic Aether Mechanics: Magnetic and Electric Fields

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Abstract

Within the Dynamic Aether Mechanics framework, the electric field is identified as the Aether flow vector—the direction and magnitude of medium flow—while the magnetic field is the flow angle—the rotational geometry or vorticity of that flow. This paper develops the electromagnetic mechanism, derives the inverse-cube falloff of magnetic dipole fields, explains ferromagnetism through Coulomb energy and Pauli exclusion, and accounts for the Meissner effect. Electromagnetic unification follows naturally: a stationary charge produces a purely radial flow vector ($\mathbf{E}$ only), but motion develops angular structure ($\mathbf{B}$ appears), with the conversion factor v^2/c^2 reflecting kinetic-to-elastic energy density. The identity $\nabla \cdot \mathbf{B} \equiv 0$ becomes a mathematical tautology $\nabla \cdot (\nabla \times \mathbf{v}) \equiv 0$, explaining why magnetic monopoles cannot exist.

This paper is part of a series presenting the Unifying Theory of Dynamic Aether Mechanics. For the foundational concepts of Aether binding, the three independent properties (density, pressure, temperature), and the unification of force constants, see Paper 1: General Overview [4].

1. Magnetism

Magnetism in this theory arises from the *angular structure* of Aether flow set in motion by electron spin. Each electron acts as a vortex, spinning the surrounding Aether into a circulatory pattern. While the electric field is the Aether flow vector—the direction and magnitude of the medium’s velocity—the magnetic field is the flow angle: the rotational geometry of that velocity field. When two or more electron vortices interact, the angular structure of their flows creates pressure differentials that produce the forces we observe as magnetism. These are Bernoulli (dynamic) pressure differentials: where opposing flows meet, kinetic energy converts to static pressure; where flows cooperate, static pressure converts to kinetic energy. This mechanism operates in the incompressible-flow limit—no density change is involved—making Bernoulli dynamics the natural description of all electromagnetic forces in a nearly incompressible superfluid. In the two-fluid model (Paper 6 [5]), electromagnetic fields are properties of the *superfluid* component: the electron vortex is a quantized circulation in the condensate, and the electric and magnetic fields are its velocity and angular flow structure. The normal component is electromagnetically nearly decoupled (coupling suppressed by

$P_{\text{th}}/K_A \sim 10^{-38}$, where $P_{\text{th}} \approx \rho_{\Lambda} c^2 \sim 6 \times 10^{-10} \text{ J/m}^3$ is the thermal pressure in cosmic voids and $K_A \sim 10^{28} \text{ J/m}^3$; see Paper 6 [5]).

1.1 The Basic Mechanism

As an electron spins, it causes the Aether around it to spin as well, creating a vortex in the surrounding medium. When two electrons come near each other, the Aether spinning around them begins to interact.

When spinning in the same direction (parallel spins), the Aether between the two electrons is impeded—the flows want to move in opposite directions at the boundary. This creates a region of high pressure between them, pushing them apart.

When spinning in opposite directions (antiparallel spins), the Aether between them flows in the same direction, allowing it to move freely and creating a region of low pressure. The surrounding higher pressure then pushes the electrons together.

The result: alike-spinning electrons repel, and opposite-spinning electrons attract. This effect is greatest at the poles of the electron, where a greater surface area of spin interacts at higher velocities, naturally producing the strongest forces along the spin axis.

Note that magnetism arises from the angular structure of interacting electron flows—the pressure differentials created at the boundary between two spinning regions of Aether. This is distinct from both the electric field (which arises from the radial Aether flow vector) and the gravitational mechanism (which arises from the vortex concentration and transient binding of Aether). All three effects originate from the electron’s interaction with Aether, but they involve geometrically distinct aspects of the same medium: the electric field is the flow vector (where and how fast the Aether moves), the magnetic field is the flow angle (how the flow pattern rotates), and gravity is a binding phenomenon (how each vortex concentrates and cycles Aether independently).

1.2 Field Topology: Closed Lines and No Monopoles

In this model, the Aether flow around each electron forms a vortex—Aether drawn inward along the spin axis, circulated outward along the equatorial plane (or vice versa), creating a continuous loop. The “magnetic field” is really a map of the angular structure of this flow pattern—the vorticity of the Aether velocity field.

This immediately explains why magnetic field lines always form closed loops. The vortex is a circulatory flow: Aether exits one pole, curves around through the surrounding space, and returns at the opposite pole. There is no source or sink—only circulation. A magnetic monopole would require a point from which Aether flow radiates outward in all directions without returning, which is not possible for a vortex. The topology of rotational flow inherently forbids monopoles.

In the flow-vector/flow-angle framework, this result is even stronger. The magnetic field is the vorticity of the Aether velocity field: $\mathbf{B} \propto \nabla \times \mathbf{v}$. The divergence of any curl vanishes identically— $\nabla \cdot (\nabla \times \mathbf{v}) \equiv 0$ —so $\nabla \cdot \mathbf{B} = 0$ is not a physical law requiring explanation but a mathematical identity. Magnetic monopoles are forbidden by the geometry of the medium, not by an empirical regularity. This is consistent with the experimental fact that no magnetic monopole has ever been observed, despite extensive searches. Every magnet, from a single electron to a superconducting coil, has two poles connected by closed field lines.

1.3 Field Falloff: The Inverse Cube Law

The magnetic field of a single electron (a magnetic dipole) falls off as the inverse cube of distance ($B \propto 1/r^3$), in contrast to gravity's inverse square law ($F \propto 1/r^2$). This difference emerges naturally from the vortex model.

Gravity is produced by the monopole effect of transient binding: each atom produces a spherically symmetric low-pressure pulse that dissipates over the surface area of an expanding sphere ($4\pi r^2$), giving $1/r^2$ falloff. The gravitational source is isotropic—it has no preferred direction.

Magnetism is produced by a dipole—a vortex with two poles and opposing flow directions. The flow pattern has both radial and angular structure. At any given distance, the flows from the two poles partially cancel (the inward flow at one pole is offset by the outward flow at the other), and this cancellation becomes more complete with distance. The net effect is an additional factor of $1/r$ beyond the geometric $1/r^2$, producing the observed $1/r^3$ falloff.

For macroscopic magnets—where the Aether flows from enormous numbers of aligned electron vortices combine—the field is more complex at close range but still follows dipole falloff at distances large compared to the magnet's size. This is consistent with observation.

1.4 Ampère's Law: Moving Charges and Magnetic Fields

A moving electric charge produces a magnetic field, as described by Ampère's law. In this theory, the mechanism connects directly to the electric field model described below: each charge produces an Aether flow vector (the electric field), and when the charge moves, that flow vector is carried along with it. The moving flow vector develops angular structure perpendicular to the direction of travel—creating the magnetic field observed around current-carrying conductors.

This reoriented flow creates a secondary rotational pattern centered on the direction of travel—exactly the circular magnetic field observed around a current-carrying wire. The faster the charge moves, the more its flow vector converts to flow angle, and the stronger the resulting magnetic field. For a wire carrying current (many moving electrons), the individual circulatory effects sum, producing a net circulatory Aether flow around the wire.

This is consistent with the right-hand rule: the orientation of the angular structure is determined by the direction of motion relative to the flow vector. The superposition of these flows produces the characteristic field pattern observed around current-carrying conductors.

1.5 Additivity and Material Permeation

Magnetism is additive. Pressure is also additive. The net sum of aligned electron spin vortices determines the total Aether flow pattern and, consequently, the total magnetic field strength. When more spins align, their vortex flows reinforce, producing proportionally stronger pressure differentials.

Magnetic fields permeate non-magnetic materials. Pressure does not require actual flowing Aether to transmit force. A non-magnetic material would disrupt the directional flow of Aether (with randomly oriented electrons each redirecting the flow), but the net sum of pressure would still propagate, creating magnetic forces beyond the material. The Aether between and around the material's atoms still transmits the pressure wave from the external vortex, just as sound pressure passes through a wall despite being scattered by the wall's structure.

1.6 Ferromagnetism: Domain Alignment and Exchange Interaction

In ferromagnetic materials (iron, nickel, cobalt), the electron spins of neighboring atoms spontaneously align within microscopic regions called magnetic domains. In the Aether model, this alignment is energetically favorable because antiparallel-spinning neighbors create low-pressure channels between them, drawing the atoms into cooperative flow patterns. Once a cluster of spins aligns, the collective vortex flow reinforces further alignment of adjacent atoms, allowing the domain to grow.

When an external magnetic field is applied, these domains reorient to align with the external Aether flow, and the material becomes strongly magnetic. When the field is removed, the domain structure can persist—producing a permanent magnet, where the frozen alignment of electron vortices maintains a self-sustaining Aether flow pattern.

The strength of ferromagnetic exchange coupling is naturally explained by the combination of two mechanisms developed elsewhere in this theory. The exchange interaction that drives ferromagnetic alignment is not an amplified magnetic dipole effect—it is fundamentally a *Coulomb energy difference* mediated by the Pauli exclusion principle.

When neighboring atoms' valence electrons have parallel spins, the topological vortex exclusion principle (Section 3) forces them into antisymmetric spatial configurations: same-spin electron vortices cannot overlap in the same spatial mode, so they remain farther apart on average. This greater separation reduces their mutual Coulomb repulsion (transmitted through radial Aether flow, Section 2), lowering the total energy. For antiparallel spins, no such topological constraint

exists—the electrons can approach more closely, increasing Coulomb repulsion.

The energy scale of this effect is set by the Coulomb interaction (~ 1 eV per atom pair), not by the magnetic dipole coupling ($\sim 10^{-4}$ eV). This explains the observed factor of $\sim 10,000$ between exchange and dipole energies without requiring any amplification mechanism—the two effects operate through entirely different Aether processes (electrostatic flow pressure versus circulatory vortex coupling). The crystal-structure dependence of ferromagnetism follows because the exchange energy depends sensitively on orbital overlap geometry: only certain lattice spacings and electronic configurations produce a net energy gain from parallel alignment.

1.7 Curie Temperature: Thermal Disruption of Alignment

A ferromagnetic material loses its magnetism when heated above a characteristic temperature (the Curie temperature). In this theory, heat is the thermal energy of Aether—the random kinetic motion of Aether particles. At low temperatures, this random motion is weak, and the organized vortex flows of aligned electron spins can maintain coherent flow patterns (magnetic domains).

As temperature rises, the thermal energy of the surrounding Aether increases. The random thermal motions of the Aether begin to disrupt the organized flow patterns, buffeting the electron vortices and breaking their alignment. At the Curie temperature, the thermal disruption is strong enough to overwhelm the cooperative flow coupling that maintains domain structure. Above this temperature, the material becomes paramagnetic—the spins can still individually respond to an external field, but they can no longer maintain spontaneous alignment against the thermal noise.

This is directly analogous to how turbulence disrupts laminar flow in classical fluids. An organized flow pattern (laminar flow / magnetic alignment) can only persist when the driving force (vortex coupling) exceeds the disrupting force (thermal turbulence). The Curie temperature marks the point of transition from ordered to disordered Aether flow.

The sharpness of this transition in many materials—where magnetism drops abruptly rather than gradually—is consistent with a cooperative phenomenon: as some spins lose alignment, the collective vortex flow weakens, making it easier for thermal motion to disrupt remaining aligned spins. The transition cascades, producing the sharp onset observed experimentally.

1.8 Paramagnetism

In paramagnetic materials, individual electron spins can partially align with an applied magnetic field, but they cannot maintain spontaneous alignment in the absence of the field. In the Aether model, this occurs because the thermal energy of the Aether is strong enough to randomize the spin orientations once the organizing external flow is removed, but not strong enough to prevent temporary alignment while the field is present.

The applied magnetic field represents an organized Aether flow that biases the orientation of electron vortices. Each vortex tends to align its axis with the external flow (because alignment reduces pressure between the vortex and the surrounding flow), but thermal fluctuations continuously knock it out of alignment. The result is partial, statistical alignment that increases with field strength and decreases with temperature—exactly as observed.

The inverse temperature dependence of paramagnetic susceptibility (Curie's law [2]) follows directly: higher temperature means more vigorous thermal disruption, less net alignment per unit field strength, and weaker paramagnetic response.

1.9 Diamagnetism

Diamagnetism is the weak repulsion that all materials exhibit in a magnetic field. It is the weakest form of magnetic response and is typically masked by stronger paramagnetic or ferromagnetic effects when present.

In the Aether model, diamagnetism arises from the response of the Aether flow within atoms when an external magnetic field (organized Aether flow) is applied. The external flow slightly modifies the existing vortex patterns of the atom's electrons. Because electrons exist in paired states with opposing spins (in filled orbitals), their vortices already produce balanced, opposing Aether flows that cancel. When an external flow is imposed, it slightly accelerates the vortex aligned with it and slightly decelerates the opposing vortex. The decelerated vortex loses less pressure to flow interaction, while the accelerated vortex gains more—the net result is a small back-pressure that opposes the applied field.

This explains why diamagnetism is universal (all atoms have electron pairs), weak (it is a second-order perturbation of already-balanced flows), and always opposing (the back-pressure always resists the external field regardless of its direction).

1.10 The Meissner Effect: Superconductivity and Complete Flux Expulsion

In superconductors, magnetic fields are completely expelled from the interior of the material (the Meissner effect). In the Aether model, this has a clear interpretation.

At superconducting temperatures, electrons form Cooper pairs—bound pairs of electrons with opposite spin. In Aether terms, each Cooper pair consists of two electron vortices spinning in opposite directions, whose flows precisely cancel. The Aether within and around the pair is, on average, stationary—no net flow, no net vortex.

When these Cooper pairs condense into a coherent superconducting state, the entire interior of the material becomes a region of zero net Aether flow. An external magnetic field (organized Aether flow) cannot penetrate this zero-flow region because there are no individual, unbalanced vortices to

couple to. The external flow is forced around the material instead, producing the observed expulsion of magnetic flux.

The characteristic penetration depth of the magnetic field into a superconductor (the London penetration depth) corresponds, in this model, to the distance over which the external Aether flow is attenuated by interaction with the Cooper pair condensate at the surface. The flow doesn't stop instantaneously at the surface—it decays exponentially over a characteristic length scale, consistent with observation.

1.11 The Electron Magnetic Moment: Antiparallel Orientation and the g -Factor

Experiments reveal that the electron's magnetic moment is antiparallel to its spin angular momentum—it points in the opposite direction. This is a quantitative constraint that the Aether model must accommodate.

In the vortex model, the spin angular momentum of the electron corresponds to the rotation of the electron itself (or its associated Aether structure). The magnetic moment, however, is determined by the circulatory Aether flow pattern the spin produces in the surrounding medium. These need not point in the same direction. If the electron's spin drives Aether inward along its spin axis and outward along the equatorial plane, the resulting circulation pattern (which determines the magnetic moment direction) would be opposite to the spin direction—consistent with the observed antiparallel relationship.

The magnitude of the electron's magnetic moment is characterized by the g -factor. The Dirac equation predicts $g = 2$ for a point particle, but the experimentally measured value is $g = 2.00231930436118$ (known to approximately thirteen significant digits [3]). The deviation from 2—the anomalous magnetic moment—arises from virtual particle interactions (transient binding events) that slightly modify the effective vortex flow pattern around the electron.

This connects directly to the transient binding mechanism central to this theory. The anomalous magnetic moment is calculated in quantum electrodynamics by summing the contributions of virtual particle loops. Each virtual pair that forms near the electron momentarily creates its own small vortex, slightly altering the net Aether flow pattern. The cumulative effect of all transient binding events is a slight enhancement of the magnetic moment beyond the bare vortex value.

The extraordinary precision with which this correction is both calculated and measured (agreement to approximately thirteen significant digits) represents both a powerful connection and a demanding constraint: any quantitative formalization of the Aether vortex model must reproduce this correction. The fact that it arises from transient binding—the same mechanism this theory proposes for gravity—suggests a deep connection between magnetic and gravitational phenomena, unified through the vortex dynamics of the electron.

2. The Electric Field: Radial Aether Flow

The electric field—the force between charged particles described by the Coulomb interaction—has a natural interpretation in the Aether framework. The electric field is identified with the *Aether flow vector*: the instantaneous velocity field of the medium, encompassing both direction and magnitude. A charged particle drives a steady radial flow of Aether—inward for one sign of charge, outward for the other—and the electric field at any point is simply the local Aether flow velocity produced by that source.

Where the electric field is the flow vector (the “what” of Aether motion—where it is flowing and how fast), the magnetic field is the *flow angle*: the rotational geometry of the Aether velocity field (the “how” of the flow pattern—how it is turning). A stationary charge produces purely radial flow with no rotational structure: $\mathbf{E}$ only, $\mathbf{B} = 0$. When that charge moves, or when the electron’s intrinsic spin drives circulation, the flow acquires angular structure—and the magnetic field appears. The magnetic field is not a separate entity but the rotational character of the same flow that constitutes the electric field.

In this model, charge is defined by the direction of the Aether flow vector. One sign of charge (negative, for the electron) corresponds to a net radial inflow of Aether toward the particle. The opposite sign (positive) corresponds to a net radial outflow. This radial flow vector is distinct from the circulatory vortex that produces the magnetic moment—it is linear, non-rotational, and spherically symmetric around the charge.

The Coulomb force follows from pressure differentials created by these radial flows, using the same mechanism that underlies both magnetism and gravity.

- Two like charges (both inflowing, or both outflowing) create a region of high Aether pressure between them—the flows compete for the same Aether, or both push Aether into the gap. This high pressure pushes the charges apart: repulsion.
- Two opposite charges (one inflowing, one outflowing) create a region of low pressure between them—the outflow of one feeds directly into the inflow of the other, creating a channel of smooth, cooperative flow. The surrounding higher pressure pushes the charges together: attraction.

This is exactly analogous to the magnetic mechanism (where parallel spins repel and antiparallel spins attract through flow-interaction pressure differentials), but operating on the flow vector rather than the flow angle. The force falls off as $1/r^2$ because the flow vector from a point source spreads over the surface area of an expanding sphere ($4\pi r^2$)—identical to the geometric falloff of gravity.

The enormous strength of the electromagnetic force compared to gravity now has a clear explanation. The Coulomb force arises from direct flow-vector pressure—the full force of the Aether flow

interacting between charges. Gravity arises from the tiny residual asymmetry of the transient bind–unbind cycle, where nearly all energy is recycled locally and only a minuscule pressure disturbance propagates outward. One is the full strength of the flow vector; the other is a faint ripple left over from a process that almost perfectly cancels itself. The ratio of electromagnetic to gravitational force (approximately 10^{42} for two electrons) reflects the ratio of direct flow-vector pressure to recycled-asymmetry pressure. Put differently, the Coulomb force is a Bernoulli effect—the full dynamic pressure of interacting Aether flows—while gravity is a compression-wave residual from the bind–unbind cycle. The 10^{42} ratio measures the difference between direct kinetic-energy-to-pressure conversion and the faint ripple of incomplete cancellation.

2.1 The Electron’s Three Aether Effects

With the electric field defined as the flow vector and the magnetic field as the flow angle, each electron now produces three distinct Aether phenomena, each from a different aspect of its interaction with the medium:

1. **Flow vector** (radial, inward for the electron)—produces the electric field. Determines charge sign and Coulomb interactions. Falls off as $1/r^2$ (spherical spreading).
2. **Flow angle** (circulatory, from spin)—produces the intrinsic magnetic moment. Determines magnetic dipole interactions with other spinning particles. Falls off as $1/r^3$ (dipole cancellation).
3. **Vortex-boundary pair creation and transient binding**—produces gravity. The vortex core acts as a rapidly moving boundary (the DCE mechanism below the permanence threshold), producing the transient bind–unbind pressure wave. Falls off as $1/r^2$ (spherical spreading of the monopole pressure wave), but is enormously weaker than the electric force because it is only the residual asymmetry of a recycling process.

These three effects coexist because they involve geometrically distinct aspects of the Aether velocity field: the flow vector (magnitude and direction of motion), the flow angle (rotational geometry), and the DCE-driven binding cycle. They are not in competition—a single electron sustains all three simultaneously.

2.2 Electromagnetic Unification: How Movement Connects E and B

The relationship between electric and magnetic fields—the foundation of Maxwell’s equations—emerges naturally from the Aether model once the electric field is identified as the Aether flow vector and the magnetic field as its angular structure.

When a charge is stationary, its Aether flow vector is purely radial—spherically symmetric, with no preferred orientation beyond the radial direction. The flow has no rotational structure, and therefore there is no magnetic field (beyond the intrinsic spin moment of the particle itself).

When a charge moves, its radial flow is carried along with it. The moving flow vector develops a circulatory component perpendicular to the direction of travel—exactly as any fluid flow develops vorticity when its source moves through the medium. This angular structure of the displaced flow is the magnetic field produced by a moving charge: movement converts part of the flow vector into flow angle.

This provides the physical mechanism behind several fundamental electromagnetic relationships:

- **Ampère’s law:** A current-carrying wire (moving charges) produces a circular magnetic field around it. In the Aether model, the moving flow vector of each electron develops angular structure from its motion through the medium, and these individual contributions sum to produce the net field around the wire.
- **The right-hand rule:** The orientation of the angular structure is determined by the direction of motion relative to the flow vector—geometry dictates the resulting flow angle.
- **Relativity of $\mathbf{E}$ and $\mathbf{B}$ fields:** In standard physics, a pure electric field in one reference frame appears as a combination of electric and magnetic fields in another frame moving relative to the first. In the Aether model, a stationary charge has a purely radial flow vector ($\mathbf{E}$ only, no angular structure). In a frame where the charge is moving, that flow vector is tilted by the motion, acquiring rotational structure ($\mathbf{B}$ appears). The Aether mechanism naturally produces the frame-dependent mixing of flow vector and flow angle that relativity describes as $\mathbf{E}$ – $\mathbf{B}$ mixing.
- **Force proportional to velocity:** The magnetic force on a moving charge ($\mathbf{F} = q\mathbf{v} \times \mathbf{B}$) depends on the charge’s velocity. In the Aether model, the pressure differential created by the interaction between two flow patterns depends on the relative motion of their sources. A stationary charge in a magnetic field (organized angular flow structure) experiences no net force because its flow vector has no velocity component to interact with the external rotation. A moving charge’s flow vector is distorted by its motion, creating asymmetric pressure interactions with the external angular structure.

2.3 Electromagnetic Induction

Faraday’s law of electromagnetic induction—a changing magnetic field produces an electric field, and a changing electric field produces a magnetic field—is the basis for generators, transformers, and electromagnetic wave propagation. The Aether model provides a physical mechanism for both directions of induction:

- **Changing $\mathbf{B}$ creates $\mathbf{E}$:** A changing magnetic field is a time-varying angular structure in the Aether flow. As this rotational pattern changes, it pushes and pulls the surrounding Aether, driving net flow—which is the electric field. The changing flow angle displaces Aether in a pattern that produces a net flow vector, inducing an electric field in the surrounding region.
- **Changing $\mathbf{E}$ creates $\mathbf{B}$:** A changing electric field is a time-varying Aether flow vector. A flow that is changing (accelerating, decelerating, or oscillating) is effectively moving through the medium, and a moving flow vector develops rotational structure—producing a magnetic field. This is Maxwell’s displacement current, given a physical interpretation: a changing flow vector acquires angular character.

2.4 Electromagnetic Waves

Together, these two processes create the conditions for self-sustaining electromagnetic wave propagation. An oscillating charge produces an oscillating flow vector (changing $\mathbf{E}$). This changing flow vector develops angular structure (creates $\mathbf{B}$). The changing angular structure in turn drives a net flow (creates $\mathbf{E}$), which changes and creates more rotation ($\mathbf{B}$), and so on. The wave propagates through the Aether as an alternating exchange between flow vector and flow angle, each regenerating the other.

This is an oscillation in the Aether medium between two aspects of the same velocity field—its magnitude/direction and its rotational geometry—propagating at the speed of light (since light speed is the propagation speed of disturbances in Aether). The electromagnetic wave does not require a separate mechanism from the other Aether phenomena—it is simply the self-sustaining oscillation between flow vector and flow angle.

The polarization of electromagnetic waves corresponds to the orientation of the flow-vector oscillation. The fact that electromagnetic waves are transverse (oscillating perpendicular to the direction of propagation) is consistent with the Aether model: the flow vector and flow angle disturbances that constitute the wave are perpendicular to each other and to the direction of propagation, just as observed.

Microscopic origin. The flow-vector and flow-angle descriptions used throughout this paper are macroscopic effective descriptions. At the microscopic level, photons are spin-wave excitations of the Aether’s spinor condensate—transverse oscillations of the internal spin orientation rather than longitudinal density perturbations (Paper 7 [6]). Through spin-orbit coupling in the condensate [7], these spin-wave modes produce effective mass currents that manifest as the velocity and vorticity fields described here. The flow-vector picture is therefore the macroscopic limit of the spin-wave dynamics, and the two descriptions are complementary rather than contradictory.

3. Summary of Theoretical Properties

3.1 Properties of Magnetism

- The magnetic field is the angular structure (flow angle) of the Aether velocity field, arising from pressure differentials between interacting electron spin vortices. This is distinct from gravity's binding mechanism. Magnetism is how the rotational geometry of two flows affects the Aether between them; gravity is how each vortex concentrates and cycles Aether independently.
- Parallel spins repel (opposing flows create high pressure between them); antiparallel spins attract (cooperative flows create low pressure between them).
- Field lines form closed loops as a topological consequence of vortex circulation. Magnetic monopoles are forbidden by a mathematical identity: $\nabla \cdot (\nabla \times \mathbf{v}) \equiv 0$.
- Dipole field falls off as $1/r^3$ (partial cancellation of opposing poles at distance), compared to gravity's monopole $1/r^2$ falloff.
- Moving charges produce magnetic fields because their flow vector (electric field) develops angular structure when carried through the medium—motion converts flow vector into flow angle (Ampère's law).
- Ferromagnetic alignment occurs when parallel-spin vortex topology forces electrons into antisymmetric spatial configurations, reducing their Coulomb repulsion. The exchange energy (~ 1 eV) is primarily a Coulomb effect mediated by Pauli exclusion (antisymmetric spatial wavefunctions reduce electron–electron repulsion), not an amplified magnetic dipole interaction.
- Curie temperature marks the transition where thermal Aether turbulence overwhelms vortex coupling, destroying domain alignment—analogue to the laminar–turbulent transition in classical fluids.
- Paramagnetic response reflects partial, statistical alignment of individual vortices against thermal disruption; diamagnetic response reflects the back-pressure from perturbation of balanced electron-pair flows.
- Superconducting flux expulsion (Meissner effect) occurs when Cooper pairs cancel all net Aether flow, creating a zero-flow interior that excludes external magnetic fields.
- The electron's magnetic moment is antiparallel to its spin, consistent with the vortex model's prediction that the circulation pattern (moment) opposes the rotation direction (spin angular momentum).

- The anomalous magnetic moment (g -factor deviation from 2) arises from transient binding events modifying the effective vortex flow pattern—connecting magnetism quantitatively to the same virtual particle mechanism that produces gravity.

3.2 Properties of Electromagnetism

- The electric field is the Aether flow vector: the instantaneous direction and magnitude of the medium's velocity at each point. Inflow for one sign of charge, outflow for the other.
- The magnetic field is the flow angle: the rotational geometry (vorticity) of the Aether velocity field. $\nabla \cdot \mathbf{B} = 0$ is a mathematical identity.
- The Coulomb force arises from pressure differentials between flow vectors: like flows create high pressure (repulsion), opposite flows create low pressure (attraction).
- The electric force falls off as $1/r^2$ (spherical spreading of the flow vector from a point source), identical in geometric origin to gravitational falloff.
- The enormous strength of the electromagnetic force relative to gravity reflects the difference between direct flow-vector pressure (electric) and the tiny residual asymmetry of the transient bind–unbind cycle (gravity).
- Each electron produces three coexisting Aether effects: flow vector (electric field), flow angle (magnetic moment), and transient binding (gravity).
- Magnetism from current arises when the flow vector (electric field) moves through the medium, developing angular structure—unifying $\mathbf{E}$ and $\mathbf{B}$ as two aspects of the same velocity field.
- Electromagnetic induction occurs because changing flow angle (changing $\mathbf{B}$) drives a net flow vector (creates $\mathbf{E}$), and changing flow vector (changing $\mathbf{E}$) develops angular structure (creates $\mathbf{B}$).
- Electromagnetic waves are self-sustaining oscillations between flow vector and flow angle, propagating at the speed of light through the medium.

3.3 Properties of Force Constants

- Only two independent electromagnetic constants exist at the Aether level: the speed of light c (wave propagation speed) and the Coulomb constant k (flow-vector pressure response).
- The magnetic permeability μ_0 is derived: $\mu_0 = 4\pi k/c^2$. It relates the flow-vector coupling to the flow-angle coupling via the medium's wave speed.

- The magnetic force between moving charges equals the electric force scaled by $v_1 v_2 / c^2$, where v_1 and v_2 are the charge velocities.
- The gravitational constant G introduces one additional parameter beyond the Aether properties in k : ϵ , the fractional asymmetry of the transient bind–unbind cycle.
- The ratio k/G decomposes into $(1/\epsilon) \times (\text{flow-to-binding ratio})^2 \times (\text{geometric factors})$, and should be derivable from Aether properties.
- The factor v^2/c^2 has a quadruple appearance: time dilation ($\gamma = 1/\sqrt{1 - v^2/c^2}$), the magnetic-to-electric force ratio ($F_{\text{mag}}/F_{\text{elec}} = v_1 v_2 / c^2$), relativistic energy ($E = \gamma m c^2$), and gravitomagnetism ($B_g = g \times v / c^2$, confirmed by GP-B). All four arise from the same Aether mechanics: v^2/c^2 is the ratio of kinetic energy density to the medium’s elastic capacity.
- The Aether is experimentally confirmed to be perfectly linear (Hookean) in its pressure–density response, confirmed by velocity time dilation (to 10^{-8} precision across velocities from $10^{-6} c$ to $0.9994c$) and independently by the Fresnel drag coefficient (exact $1 - 1/n^2$ across materials with density ratios up to nearly 6:1). This compressive linearity governs wave propagation; the electromagnetic forces in this paper arise from Bernoulli pressure in the incompressible-flow limit.
- Gravitational time dilation ($\Delta t/t = GM/(rc^2)$) directly connects G to the medium: the Aether inflow velocity $v_{\text{in}} = \sqrt{2GM/r}$ produces $v_{\text{in}}^2/(2c^2) = GM/(rc^2)$, the time dilation of a static observer in the flowing medium.
- The fine structure constant ($\alpha \approx 1/137$) measures the strength of a single electron’s radial flow relative to the quantum scale of the medium.
- The gravitational coupling constant ($\alpha_G \approx 1.75 \times 10^{-45}$) measures the strength of a single electron’s residual binding pressure relative to the same quantum scale.
- The Planck mass ($m_P \approx 2.18 \times 10^{-8}$ kg) represents the mass scale at which gravitational binding pressure becomes comparable to quantum-scale Aether disturbances.

4. Experimental Support and Connections to Established Physics

The following experimentally confirmed phenomena support or align with the aspects of the theory presented in this paper:

1. **The anomalous magnetic moment of the electron.** Virtual particle corrections to the electron’s magnetic moment have been confirmed to approximately thirteen significant digits [3],

establishing that transient particle–antiparticle formation and annihilation is a real, measurable process occurring continuously around charged particles. In this theory, the anomalous moment arises from transient binding events (the same mechanism that produces gravity) modifying the effective Aether vortex flow around the electron—connecting the magnetic and gravitational mechanisms through a single, precisely measured quantity.

2. **Absence of magnetic monopoles.** Despite extensive experimental searches, no magnetic monopole has ever been observed. This is a direct consequence of the vortex model: circulatory Aether flow inherently produces closed field lines with two poles. A monopole would require a topologically impossible radial-only flow from a vortex.
3. **The Meissner effect.** Superconductors expel magnetic fields from their interiors. In this theory, Cooper pairs (opposite-spin electron pairs) cancel all net Aether flow, creating a zero-flow region that external magnetic fields cannot penetrate. The observed exponential decay of the field at the surface (London penetration depth) is consistent with the gradual attenuation of external Aether flow by the Cooper pair condensate.
4. **The $\mu_0 = 4\pi k/c^2$ relationship.** Maxwell’s equations [1] establish that the speed of light equals $1/\sqrt{\epsilon_0\mu_0}$, linking the electric and magnetic constants through c . In the Aether model, this relationship is not a mathematical coincidence but a physical necessity: magnetism is moving electric field, so the magnetic constant must equal the electric constant divided by the square of the propagation speed.

5. Open Questions and Testable Predictions

The following areas relevant to this paper remain underdeveloped and represent opportunities for further refinement or falsification:

1. **Quantitative reproduction of the g -factor anomaly.** The electron’s anomalous magnetic moment ($g = 2.00231930436118$) is known to extraordinary precision and is calculated in QED by summing virtual particle loop corrections. If the Aether vortex model can reproduce this correction from first principles—by calculating how transient binding events modify the effective vortex flow pattern—it would provide one of the most stringent quantitative tests of the theory.
2. **Origin of charge sign.** If charge corresponds to the direction of radial Aether flow (inflow vs. outflow), what determines which direction a given particle adopts? The electron’s inward vortex (drawing Aether toward its center) suggests a natural connection between the spin vortex direction and charge sign. The existence of positrons (positive electrons with opposite charge

but equal mass) suggests that the flow direction can be reversed while preserving the binding properties that determine mass.

3. **Deriving the fine structure constant from Aether properties.** The fine structure constant ($\alpha \approx 1/137$) measures the coupling strength of a single electron's radial flow to the quantum scale of the medium. If the electron's radial flow rate (ϕ) can be calculated from the vortex geometry, and the quantum scale of Aether disturbances is set by $\hbar$ and c , then $\alpha = ke^2/(\hbar c)$ should be derivable from first principles.

6. Mathematical Summary

This section collects the key equations relevant to the topics covered in this paper. *For the complete mathematical summary, see Paper 1: General Overview [4].*

6.1 Fundamental Aether Parameters

The theory posits four microscopic parameters from which all macroscopic phenomena derive:

Symbol	Name	Constrained range	Primary constraint
ρ_A	Ambient density	$10^{11}\text{--}10^{12} \text{ kg/m}^3$	Electron vortex energy
K_A	Bulk modulus	$10^{28}\text{--}10^{29} \text{ J/m}^3$	$K_A = \rho_A c^2$
m_A	Aether quantum mass	$500\text{--}850 \text{ MeV}/c^2$	Flux tube width / glueball mass
ξ	Healing length	$0.17\text{--}0.28 \text{ fm}$	Bridge diameter (lattice QCD)

These four are not independent. Two defining relations connect them:

$$c = \sqrt{K_A/\rho_A} \quad (1)$$

$$\xi = \frac{\hbar}{m_A c \sqrt{2}} \quad (2)$$

leaving two free Aether parameters (e.g., ρ_A and m_A) from which all others follow. (A third parameter, ϵ , enters through gravity; see below.)

6.2 Equation of State

The Aether obeys a logarithmic equation of state, the unique form that produces a density-independent bulk modulus (Section 4):

$$P = K_A \ln\left(\frac{\rho}{\rho_0}\right) + P_0 \quad (3)$$

Key consequence: the speed of sound $c_s = \sqrt{K_A/\rho}$ varies with density. At ambient density, $c_s = c$. Note: in gravitational fields, Bernoulli's equation for free-fall inflow gives $\Delta\rho/\rho_A = 0$ exactly, so c does not vary near gravitating masses; the density dependence is relevant for material media and for the two-fluid cosmological structure. The electromagnetic forces described in this paper arise from Bernoulli (dynamic) pressure—the redistribution of kinetic and static energy in flowing Aether—not from this compressive equation of state. The equation of state governs wave propagation and cosmological structure; electromagnetic mechanisms require only the incompressible-flow limit. This equation of state guarantees that K_A is constant under compression, which is experimentally confirmed by:

- Velocity time dilation measurements spanning $10^{-6} c$ to $0.9994 c$ (precision $\sim 10^{-8}$).
- The Fresnel drag coefficient $1 - 1/n^2$, exact across materials with density ratios up to $\sim 6:1$.

6.3 Electromagnetic Equations

Electric field (flow vector). The electric field is the Aether flow vector—the instantaneous direction and magnitude of the medium's velocity. The Coulomb force between charges q_1, q_2 at separation r :

$$F_{\text{elec}} = k \frac{q_1 q_2}{r^2} \quad (4)$$

Magnetic force (flow angle). Magnetism arises from the angular structure of the Aether flow—the rotational geometry of the velocity field. The magnetic force between charges moving at velocities v_1, v_2 :

$$F_{\text{mag}} = F_{\text{elec}} \frac{v_1 v_2}{c^2} \quad (5)$$

Permeability. The magnetic permeability relates the flow-vector coupling to the flow-angle coupling, and is a derived quantity, not a fundamental constant:

$$\mu_0 = \frac{4\pi k}{c^2} \quad (6)$$

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